

Solution Architecture

Date	15 March 2026
Team ID	-
Project Name	Rising Waters - Flood Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram (Text)

Presentation Layer: HTML, CSS, JavaScript Web UI

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API Gateway: Flask Routes (/ and /predict)

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Auth Service: Basic Flask Request Handling

Core Logic: Feature Engineering + XGBoost Model + Prediction

External Integration: Kaggle CSV Dataset

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Data Layer: flood.csv dataset + flood_model.pkl

Component Description Table

Component	Description	Technologies
Presentation Layer	Collects flood parameters and displays prediction.	HTML, CSS, JavaScript
API Gateway	Receives requests and returns JSON predictions.	Flask
Core Logic Service	Feature engineering, XGBoost inference, probability calculation.	Python, Pandas, XGBoost, Pickle
Data Storage	Stores training dataset and trained model.	Kaggle CSV (flood.csv), flood_model.pkl